Lesson 12 Teacher's Key: Hearing It Happen

TIC-80 - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

No new TIC-80 mechanism this week - sfx() was already fully established by Lesson 2, and the Lesson 8 slots (lifelost, gameover) already exist. This lesson adds one more slot (wave) in the exact same style and makes the gameover slot mandatory if it was skipped.

Level 1 - Full Worked Solution

- sfx(6) (wave) is the one genuinely new sound this week, placed exactly where Lesson 11's nested if already lives.
- sfx(5) (gameover), optional since Lesson 8, becomes mandatory starting today - if a student skipped it, this is the lesson where it has to go in.
- sfx(2) (hit) stays exactly where it was (inside the bullets loop's hit branch) - this lesson only touches the wave-advance and escape branches.

Why This Structure

- TIC-80's collision detection scans every frame, but a hit only ever causes enemies_remaining to cross to a new value once - so code placed directly inside that specific crossing (not a separately-polled check in TIC() itself) is naturally edge-triggered, with no manual guard needed.
- Keeping every sound call physically adjacent to the game-state change it represents (wave advance, then immediately the matching sfx() call) keeps this lesson's addition easy to audit later.

Level 2 / Level 3 Talking Points

- Ask students to name all six SFX Editor slots and their triggers from memory as a quick comprehension check before moving to lab time.
- For advanced students: ask what would need to change to give the hit sound slight pitch variation using math.random() - a natural preview of expanding beyond fixed sfx() calls later in the year.

Common Student Mistakes

- Placing sfx(6) outside the else branch, causing it to also fire on the true win.
- Mixing up slot numbers between adjacent slots - easy to do since they were designed close together in the SFX Editor.

- Skipping sfx(5) (gameover) because it was optional before - it's mandatory as of this lesson.